

Statistical Modeling (MT-2002)

Date: Nov 5, 2025

Course Instructors

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Sessional II Exam

Total Time (Hrs): 1

Total Marks: 55

Total Questions: 3

Roll No

Section

Student Signature

Attempt all the questions

Use answer sheet to answer question 1 and 2

Question 3 should be attempted on bubble sheet attached

DO NOT WRITE BELOW THIS LINE

CLO # 4: Assess the suitability of a model given the data.

Question # 1

A data analyst is working with a customer dataset that contains several predictors and two outcome variables. A small sample of the data is shown below:

CustomerID	x_1 (Age)	x_2 (Income)	x_3 (Visits)	y_1 (Annual Spending)	y_2 (Renewed)
1	25	45,000	8	530.2	1
2	31	58,500	12	610.7	1
3	47	73,200	5	420.5	0
4	29	39,800	7	480.9	1
5	55	82,600	3	395.0	0

Here, y_1 represents a **continuous** variable (spending in USD), while y_2 represents a **binary** variable indicating subscription renewal.

Two Bayesian models are proposed for analysis:

Model 1

Model 2

$$\alpha \sim \mathcal{N}(0, 10)$$

$$\beta \sim \mathcal{N}(0, 5, \text{shape} = 3)$$

$$\sigma \sim \text{HalfCauchy}(0, 5)$$

$$\mu_i = \alpha + \beta x_{1i}$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

$$\alpha \sim \mathcal{N}(0, 10)$$

$$\beta \sim \mathcal{N}(0, 5, \text{shape} = 3)$$

$$p_i = \text{logit}^{-1}(\alpha + \beta x_{1i})$$

$$bd = -\alpha/\beta$$

$$y_i \sim \text{Bernoulli}(p_i)$$

Questions:

- (a) Based on the model specifications above, identify which model (**Model 1** or **Model 2**) is more appropriate for:

- (a)i. Predicting a **continuous** target variable (y_1)

Model 1 assumes a Normal likelihood $y_i \sim \mathcal{N}(\mu_i, \sigma)$, which is appropriate for predicting the continuous target variable y_1 .

- (a)ii. Predicting a **binary** target variable (y_2)

Model 2 assumes a Bernoulli likelihood $y_i \sim \text{Bernoulli}(p_i)$ with a logistic link, which is appropriate for predicting the binary target variable y_2 .

Briefly justify your selection in one line for each.

[2 Marks]

- (b) Now, consider updated versions of **Model 1** and **Model 2** that include additional predictor variables ($x_1, x_2, \dots, x_n$).

- (b)i. For the revised **Model 1**, what potential issues might arise as more predictors are added, why could these occur, and how can they be addressed effectively? *Hint: Refer to concepts discussed in Chapter 3.* [9 Marks]

Confounding variables: A confounder influences both the predictors and the outcome, leading to biased coefficient estimates. *Mitigation: include known confounders or use hierarchical/partial pooling to reduce bias.*

Multicollinearity: Predictors are highly correlated, making coefficient estimates unstable and inflating variances. *Mitigation: use regularisation (ridge or Bayesian shrinkage priors) or remove redundant predictors.*

Masking effect: A predictor appears insignificant because other correlated variables hide its influence. *Mitigation: perform variable selection, check predictor correlations, or use dimension reduction methods.*

- (b)ii. For the revised **Model 2**, what challenges may occur when multiple predictors are included, why might they happen, and how could they be mitigated? *Hint: Relate your answer to ideas introduced in Chapter 4.* [9 Marks]

High collinearity: Strong correlation among predictors distorts logistic regression coefficients and reduces model interpretability. *Mitigation: apply shrinkage priors or remove/merge highly correlated features.*

Class imbalance: One class (e.g., renew vs not renew) dominates, causing biased probability estimates and poor minority class prediction. *Mitigation: use balanced sampling, informative priors, or weighted likelihoods.*

[Total: 20 Marks]

CLO # 2: Fit models to data using computational tools available in Python.

Question # 2 Suppose x_1 alone is not sufficient to explain the variation in the data for both regression and classification tasks. Modify Model 1 and Model 2 to include an additional predictor x_2 , rewriting each model in statistical notations accordingly for two variables. [5+5 Marks]

Model 1 (Updated: Two Predictors)

Model 2 (Updated: Two Predictors)

$$\alpha \sim \mathcal{N}(0, 10)$$

$$\beta_1 \sim \mathcal{N}(0, 5)$$

$$\beta_2 \sim \mathcal{N}(0, 5)$$

$$\sigma \sim \text{HalfCauchy}(0, 5)$$

$$\mu_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i}$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

$$\alpha \sim \mathcal{N}(0, 10)$$

$$\beta_1 \sim \mathcal{N}(0, 5)$$

$$\beta_2 \sim \mathcal{N}(0, 5)$$

$$p_i = \text{logit}^{-1}(\alpha + \beta_1 x_{1i} + \beta_2 x_{2i})$$

$$y_i \sim \text{Bernoulli}(p_i)$$

$$\text{bd: } x_2 = -\frac{\alpha + \beta_1 x_1}{\beta_2} \quad (\text{if } \beta_2 \neq 0)$$

Question # 3

[25 Marks]

1. Which of the following best defines regression in statistical modeling?
 - (a) Modeling random variables
 - (b) Estimating the relationship between dependent and independent variables
 - (c) Calculating the mean of independent variables
 - (d) Assessing correlation between data points
2. What does the slope coefficient represent in a simple linear regression model?
 - (a) The variance of residuals
 - (b) The intercept of the regression line
 - (c) The change in the dependent variable for a one-unit change in the independent variable
 - (d) The mean of the dependent variable
3. In robust linear regression, which distribution is typically used to handle outliers?
 - (a) Gaussian Distribution
 - (b) Binomial Distribution
 - (c) Student's t-distribution
 - (d) Exponential Distribution
4. Which of the following correctly matches the appropriate likelihood function to the given data type in a Bayesian PyMC model?
 - (a) Normal for binary classification, Gamma for count data, Student-T for heavy-tailed continuous data, Bernoulli for strictly positive data.
 - (b) Normal for symmetric continuous data, Gamma for strictly positive skewed data, Student-T for heavy-tailed continuous data, Bernoulli for binary data.
 - (c) Normal for positive skewed data, Gamma for binary data, Student-T for symmetric continuous data, Bernoulli for count data.
 - (d) Normal for heavy-tailed data, Gamma for symmetric continuous data, Student-T for binary classification, Bernoulli for count data.
5. The determination coefficient R^2 measures:
 - (a) The variance of residuals
 - (b) The proportion of variance in the dependent variable explained by the independent variables
 - (c) The strength of the relationship between predictors
 - (d) The goodness of fit for logistic models
6. Multicollinearity in multiple linear regression implies:
 - (a) Independent variables are uncorrelated
 - (b) Predictors have constant variance
 - (c) Predictors are highly correlated, causing instability in coefficient estimation
 - (d) There is a perfect linear relationship
7. What parameter do we estimate through simple linear model?
 - (a) probabilities

- (b) only slope
 - (c) only intercept
 - (d) slope and intercept.
8. What role does the parameter ν play in a robust linear regression with a Student's t-distribution?
- (a) It controls the intercept value
 - (b) It adjusts the effect of outliers
 - (c) It defines the slope steepness
 - (d) It measures residual variance
9. Anscombe's Quartet primarily demonstrates:
- (a) The impact of multicollinearity
 - (b) The importance of data visualization and outlier detection
 - (c) The effect of autocorrelation
 - (d) The challenges of polynomial regression
10. Anscombe's Quartet has the following number of sub-datasets :
- (a) 6
 - (b) 12
 - (c) 11
 - (d) 4
11. In a Bayesian PyMC model, when should a Bernoulli distribution be used as a likelihood function instead of a Normal distribution?
- (a) When modeling continuous data with symmetric variability around a mean.
 - (b) When modeling binary outcomes, such as success/failure or 0/1 responses.
 - (c) When modeling count data, such as the number of occurrences in a fixed interval.
 - (d) When modeling strictly positive, skewed data like waiting times.
12. In a Bayesian PyMC model, when choosing a likelihood function, under which scenario is a Gamma distribution preferred over a Normal distribution?
- (a) When modeling data that can take both positive and negative values.
 - (b) When modeling positive-only skewed data, such as waiting times or rates.
 - (c) When modeling symmetric data centered around a mean.
 - (d) When modeling data with normally distributed errors.
13. Which of the following is a confounding variable?
- (a) A variable correlated with the dependent variable only
 - (b) A variable correlated with both dependent and independent variables
 - (c) A variable that has no influence on the dependent variable
 - (d) An irrelevant variable in the model
14. The masking effect occurs when:
- (a) Residuals have constant variance

- (b) Outliers distort the model fit
 - (c) The effect of one predictor hides the impact of another
 - (d) Predictors have low correlation
15. What does it mean when the spread of errors (residuals) changes as the value of the predictor variable increases?
- (a) The relationship between the variables is perfectly linear.
 - (b) The prediction errors have the same spread for all predictor values.
 - (c) The prediction errors become larger or smaller depending on the predictor value.
 - (d) Multicollinearity among predictors
16. Posterior predictive checks (PPC) in Bayesian regression are used to:
- (a) Validate model assumptions
 - (b) Standardize predictors
 - (c) Minimize residual variance
 - (d) Adjust hyperparameters
17. Which metric measures the linear relationship between two continuous variables?
- (a) R-squared
 - (b) Mean Absolute Error
 - (c) Pearson correlation coefficient
 - (d) Log-Likelihood
18. When would you prefer using a pooled (hierarchical) model?
- (a) When predictors are orthogonal
 - (b) When analyzing completely independent groups
 - (c) When groups are assumed to share similarities
 - (d) When multicollinearity is severe
19. Why is the Pearson correlation inappropriate for non-linear relationships?
- (a) It assumes heteroscedasticity
 - (b) It measures only linear associations
 - (c) It ignores the intercept of the model
 - (d) It requires normally distributed data
20. In a Bayesian regression approach, a wide posterior distribution for a slope coefficient suggests:
- (a) Multicollinearity
 - (b) High uncertainty in the parameter estimate
 - (c) A strong predictor variable
 - (d) Overfitting of the model
21. Why use synthetic data in linear regression modeling?
- (a) To estimate multicollinearity
 - (b) To validate model assumptions under controlled scenarios

- (c) To optimize hyperparameters
 - (d) To minimize residual variance
22. Which of the following statements about the Pearson correlation coefficient r is **incorrect**?
- (a) r always lies between -1 and 1 .
 - (b) A correlation of $r = 0$ implies that X and Y are independent.
 - (c) The square of the Pearson correlation coefficient, r^2 , represents the proportion of variance in Y explained by X .
 - (d) Pearson correlation captures only linear relationships between variables.

For remaining MCQs Consider the following PyMC models

- (a)

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", mu=0, sigma=1)
    beta = pm.Normal("beta", mu=0, sigma=1)
    epsilon = pm.HalfCauchy("epsilon", beta=1)
    mu = alpha + beta * X
    Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
```
- (b)

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", mu=0, sigma=1)
    beta1 = pm.Normal("beta1", mu=0, sigma=1)
    beta2 = pm.Normal("beta2", mu=0, sigma=1)
    epsilon = pm.HalfCauchy("epsilon", beta=1)
    mu = alpha + beta1 * X1 + beta2 * X2
    Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
```
- (c)

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", mu=0, sigma=1)
    beta1 = pm.Normal("beta1", mu=0, sigma=1)
    beta2 = pm.Normal("beta2", mu=0, sigma=1)
    epsilon = pm.HalfCauchy("epsilon", beta=1)
    mu = alpha + beta1 * X + beta2 * X**2
    Y_obs = pm.Normal("Y_obs", mu=mu, sigma=epsilon, observed=Y)
```
- (d)

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", mu=0, sigma=1)
    beta = pm.Normal("beta", mu=0, sigma=1)
    v = pm.Exponential("v", lam=1)
    mu = alpha + beta * X
    Y_obs = pm.StudentT("Y_obs", mu=mu, sigma=1, nu=v, observed=Y)
```

23. Which of the above mentioned PyMC models correctly represents polynomial greater than degree 1 regression?

Options: a, b, c, d

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24. Which one of the mentioned above PyMC models represents a simple linear regression model?

Options: a, b, c, d

25. Which one of the above mentioned PyMC models represents a multiple linear regression model?

Options: a, b, c, d

Q No	Correct
Subject 1 - .	
1	B
2	C
3	C
4	B
5	B
6	C
7	D
8	B
9	B
10	D
11	B
12	B
13	B
14	C
15	B
16	A
17	C
18	C
19	B
20	B
21	B
22	B
23	C
24	A
25	B

Figure 1: Question 3